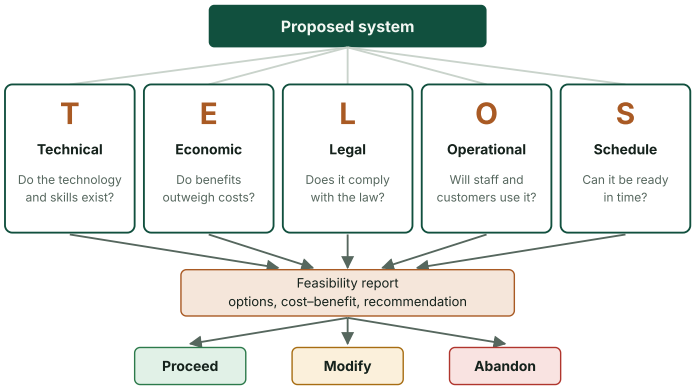
User needs, time and cost: the feasibility study and the development team

AP02 FACT FILE

Specification: students should be able to

- Explain the need for software systems that meet the needs of organisations and/or individuals
- Explain the main factors affecting systems development: the user needs, time and cost
- Describe the purpose of a feasibility study and explain technical, economic, legal, operational and schedule feasibility
- Explain the use of a cost–benefit analysis and describe the content of a feasibility report
- Describe the roles of the systems analyst, the project manager and the programmer during systems development

Key words: user needs, constraint, budget, deadline, feasibility study, TELOS, technical feasibility, economic feasibility, legal feasibility, operational feasibility, schedule feasibility, cost–benefit analysis, tangible benefit, intangible benefit, payback period, feasibility report, systems analyst, project manager



A feasibility study tests a proposed system against five TELOS criteria and produces a feasibility report recommending to proceed, modify or abandon

Meeting needs within time and cost

Every new system is developed to meet a need, and it must satisfy two groups at once. The organisation needs the system to fulfil its purpose: a ferry operator, for example, cannot take thousands of bookings a week by phone, so an online system is essential. The individuals who use the system, whether customers or staff, need it to be accessible, quick and easy to use; if customers find it confusing they will go to a competitor, and if staff find it awkward they will make mistakes or find workarounds, so the system fails. Development is also shaped by three factors: the user needs, the time available and the cost. User needs define what must be built, time sets the deadline by which it must work, and cost sets the budget. Before committing to a project, an organisation carries out a feasibility study to check that the proposed system can meet its needs within these limits, and a team of specialists, led by the systems analyst, project manager and programmer, then carries the work through. This lesson covers each of these ideas.

User needs, time and cost as constraints

The user needs are the requirements that the organisation and its users have of the system: what it must do, what data it must handle and how people will interact with it. They must be identified precisely and agreed, because a misunderstood requirement leads to the wrong system being built. Time is a constraint because most systems have a fixed date by which they must be operational, such as the start of a financial year, a tourist season or a change in the law. There is no value in designing a feature that will take a year to build if the system must go live in six months, so the scope must fit the schedule. Cost is a constraint because every project has a budget. The budget limits how many developers can be employed, what hardware and software can be bought, and how much training and testing can be paid for. The three factors interact: adding features increases time and cost, a shorter deadline needs more staff or fewer features, and a smaller budget means a smaller system. Cutting testing to save time or money is a false economy that echoes the failures of the software crisis.

Purpose of a feasibility study and TELOS

A feasibility study is a short, preliminary investigation that decides whether a proposed system should be developed. It examines the problem, looks at alternative solutions, including improving or keeping the current system, and evaluates them against five criteria often remembered as TELOS. Technical feasibility asks whether suitable hardware, software, network capacity and technical skills exist. Economic feasibility weighs costs against benefits. Legal feasibility checks compliance with legislation such as the UK GDPR and Data Protection Act 2018, the Computer Misuse Act 1990 as amended, copyright law and accessibility duties. Operational feasibility considers whether the system will fit the organisation's working practices and whether staff and customers will accept and use it. Schedule feasibility asks whether it can be delivered by a required deadline. The output is a feasibility report containing the terms of reference, the problems with the current system, the options considered, the TELOS findings with outline costs and benefits, and a justified recommendation with an outline plan. Management use it to decide whether to approve and fund the project, so it reduces the risk of committing large sums to an unworkable idea.

Cost–benefit analysis and payback

A cost–benefit analysis lists the one-off and ongoing costs of a proposed system and compares them with its benefits over a period, often three to five years. One-off costs include development, hardware, software licences, converting existing data and initial training; ongoing costs include maintenance, support, licence renewals and consumables. Tangible benefits can be given a money value, such as fewer staff hours spent on data entry or reduced stock wastage, while intangible benefits, such as improved reputation, better customer satisfaction or better decision-making, are harder to quantify but still matter. The payback period is the time taken for cumulative benefits to equal cumulative costs. To calculate it, set out the costs and benefits year by year, work out the net figure for each year and keep a running cumulative total; payback occurs when the total reaches zero. For example, a £30 000 system with running costs of £4 000 a year that saves £14 000 a year has a net saving of £10 000 a year, so it pays back in 30 000 ÷ 10 000 = 3 years. Because every figure is an estimate, the analysis should state its assumptions.

The systems analyst

The systems analyst's central job is to establish what users need from a new or changed computer system and what data processing it must carry out. The analyst acts as the link between the client, who understands the business, and the technical team, who will build the system. Early in a project the analyst carries out the feasibility study and writes the feasibility report. If the project goes ahead, the analyst investigates the current system using fact-finding methods: interviewing managers and users, observing staff at work, issuing questionnaires to large groups and inspecting existing documents such as forms, invoices and reports. The findings are analysed to establish exactly what the new system must do, and the analyst then defines the system specification, the detailed proposal for a new, modified or replacement system that the programmers will work from. Throughout, the analyst liaises with the client to make sure requirements are understood and agreed. Good analysts need strong communication skills as well as technical knowledge, because a requirement that is misunderstood at this stage becomes very expensive to correct once code has been written.

The project manager and the programmer

The project manager plans the project, co-ordinates the people working on it and keeps control of its spending. They make sure it is completed on time, within budget and to its objectives, that resources are used efficiently and that everyone involved is doing their job. Typical duties are planning the schedule, splitting the project into tasks and sub-tasks, allocating staff, hardware and software to each task, identifying risks, monitoring progress against milestones and reporting back to the client. The programmer builds and tests the system to the analyst's design. Programmers write or amend source code in a high-level language, working from module specifications, algorithms or flowcharts. They comment their code meaningfully so others can maintain it, test it using a test plan, debug any faults found, integrate the new code with existing systems, write operational documentation and work closely with the analyst and project manager so that requirements are met in the expected time. A programmer who built a system is well placed to maintain it later, and experienced programmers are valuable for maintaining and enhancing other systems too.

T Key terms from the mark schemes

Feasibility study
A preliminary investigation to decide whether a proposed system is technically, economically, legally, operationally and schedule feasible, ending in a recommendation.
Credited: TELOS

Feasibility report
The document produced by a feasibility study that describes the options considered, their feasibility and a justified recommendation for management.

Operational feasibility
Whether a proposed system will work in practice in the organisation and be accepted and used by its staff and customers.

Cost–benefit analysis
A comparison of the estimated costs of a proposed system with its expected tangible and intangible benefits over a period of time.

Payback period
The length of time it takes for the cumulative benefits of a system to equal its cumulative costs.

Intangible benefit
A benefit that is difficult to measure in money, such as improved customer satisfaction or staff morale.

Tangible benefit
A benefit that can be measured in money, such as reduced staffing or postage costs.

User needs
The requirements that an organisation and the individuals who use a system have of it: what it must do and how they need to interact with it.
Credited: User requirements

Systems analyst
The person who identifies the user and data processing requirements of a new or modified system, carrying out the feasibility study, fact-finding and producing the system specification.

Project manager
The person responsible for planning, co-ordinating and controlling a project so that it is delivered on time, within budget and meets its objectives.

Programmer
The person who writes, documents, tests and debugs the source code for the system designed by the systems analyst.

C C# example

Programmer

```
// Module: payback period (spec 4.2 from the analyst)
// Input: one-off cost and yearly net saving in pounds
// Output: number of whole years until costs are recovered
double initialCost = 30000;      // development + hardware
double netSavingPerYear = 10000; // savings minus running costs
double cumulative = -initialCost;
int year = 0;

// Add each year's net saving until the total is no longer negative
while (cumulative < 0)
{
    year++;
    cumulative += netSavingPerYear;
}
Console.WriteLine($"Payback reached in year {year}");
```

Exam focus

When asked to explain an area of feasibility, name it and then apply it to the scenario, for example: Operational, the staff are not confident with computers, so they may resist the change unless they are trained (2). Do not simply define the word economic. For calculation questions, set out costs and benefits year by year with a cumulative total; method marks are often available even if the final figure is wrong. For roles, questions typically ask for the main role (1 mark) and a number of duties (1 mark each). Mark schemes credit the analyst identifying user and data processing requirements, doing the feasibility study, liaising with the client, fact-finding and defining the system specification; the project manager planning the schedule, splitting tasks, allocating resources, identifying risks, monitoring progress, keeping to time and budget, and reporting to the client; and the programmer coding from specifications, commenting, debugging, testing with a test plan, integrating with existing systems and writing documentation. A common mistake is giving the programmer the analyst's fact-finding duties, or the project manager the coding. For needs, time and cost, explain the effect on the project, not just the factor's name.

? Check yourself

1. State the purpose of a feasibility study. [1]
2. A ferry company operating from Larne is developing an online booking system. Explain why the system must meet the needs of both the company and its individual customers. [4]
3. Explain how time and cost can each limit what is delivered when a new system is developed for a Belfast secondary school. [4]

Answers: 1. To decide whether a proposed system is possible/worthwhile/should go ahead, resulting in a recommendation to management (1) 2. Company (max 2): the volume of bookings is too large to handle by phone or in person (1), so an online system is needed to take bookings and payments efficiently and fill sailings (1) / Customers (max 2): customers must find it accessible and easy to use, e.g. clear choice of sailing, vehicle type, payment (1); if not, they may book with a competitor so the system fails its purpose (1) 3. Time (2): the system must be operational by a fixed date, e.g. the start of the school year (1), so features that cannot be built and tested by then must be dropped or delayed (1) / Cost (2): the budget is limited (1), so it restricts the number of developers, hardware purchased, licences, training or amount of testing (1)